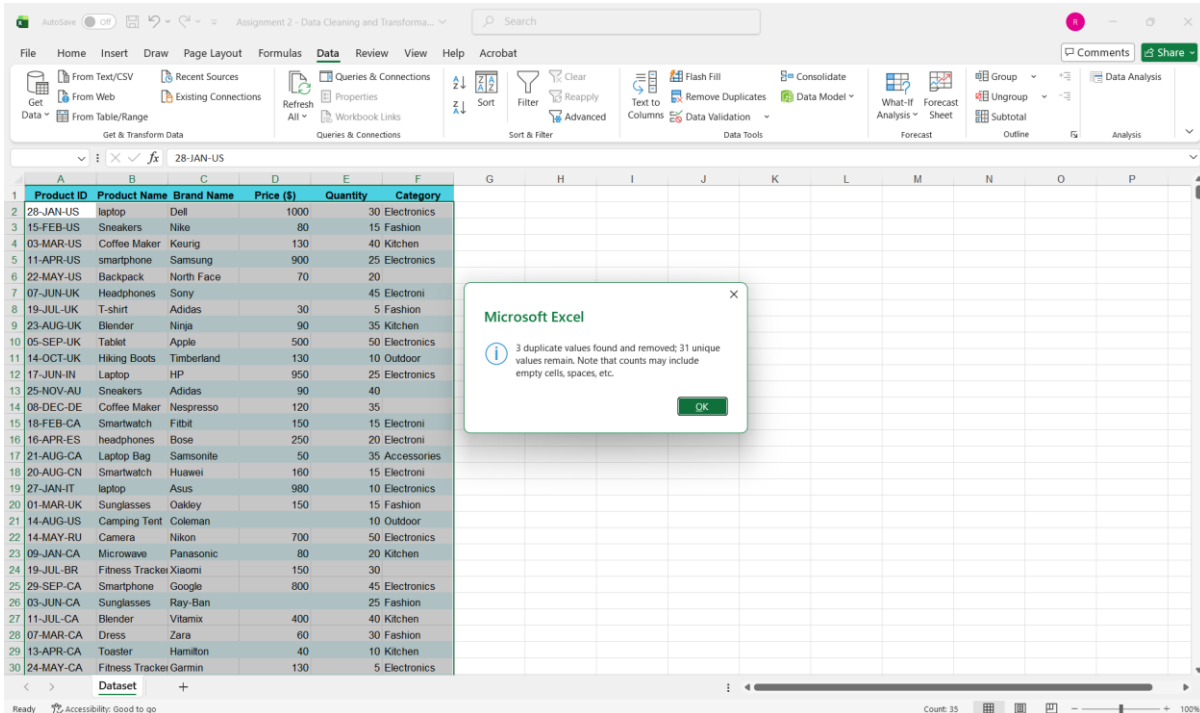
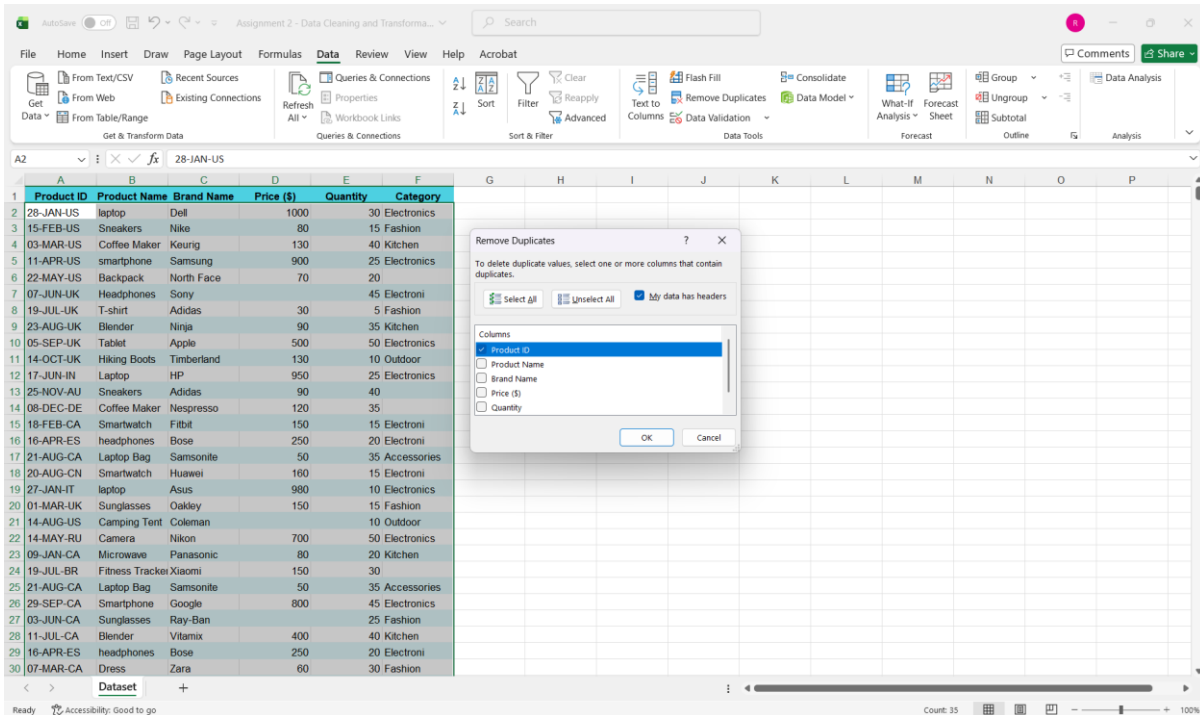


Excel Assignment 2

Step1: remove duplicates

I removed duplicate records from the given data set based on “product id” using the data – Remove duplicates option in excel



Step 2: correcting inconsistent data

I corrected the product names and category data in the data set using trim and proper functions, I used the find and replace option to correct the spelling errors.

The screenshot displays the Microsoft Excel interface with a data table and a 'Find and Replace' dialog box open.

Excel Interface:

- File Name:** Assignment 2 - Data Cleaning and Transforma...
- Formulas Tab:** Active, showing options like 'Queries & Connections', 'Refresh All', and 'Workbook Links'.
- Home Tab:** Active, showing options like 'Sort', 'Filter', 'Clear', 'Reapply', 'Text to Columns', 'Remove Duplicates', 'Data Validation', 'Flash Fill', 'Consolidate', 'Data Model', 'What-If Analysis', 'Forecast Sheet', 'Group', 'Ungroup', 'Subtotal', 'Outline', and 'Analysis'.
- Worksheet:** 'Electroni' (Sheet1).
- Table:** A table with 4 columns: Product ID, Product Name, Brand Name, and Category. The data is as follows:

Product ID	Product Name	Brand Name	Category
28-JAN-US	Laptop	Dell	Electronics
15-FEB-US	Sneakers	Nike	Fashion
03-MAR-US	Coffee Maker	Keurig	Kitchen
11-APR-US	Smartphone	Samsung	Electronics
22-MAY-US	Backpack	North Face	
07-JUN-UK	Headphones	Sony	Electroni
19-JUL-UK	T-Shirt	Adidas	Fashion
23-AUG-UK	Blender	Ninja	Kitchen
05-SEP-UK	Tablet	Apple	Electronics
14-OCT-UK	Hiking Boots	Timberland	Outdoor
17-JUN-IN	Laptop	Hp	Electronics
25-NOV-AU	Sneakers	Adidas	
08-DEC-DE	Coffee Maker	Nespresso	
18-FEB-CA	Smartwatch	Fitbit	Electroni
16-APR-ES	Headphones	Bose	Electroni
21-AUG-CA	Laptop Bag	Samsonite	Accessories
20-AUG-CN	Smartwatch	Huawei	Electroni
27-JAN-IT	Laptop	Asus	Electronics
01-MAR-UK	Sunglasses	Oakley	Fashion
14-AUG-US	Camping Tent	Coleman	Outdoor
14-MAY-RU	Camera	Nikon	Electronics
09-JAN-CA	Microwave	Panasonic	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	
29-SEP-CA	Smartphone	Google	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	Fashion
11-JUL-CA	Blender	Vitamix	Kitchen
07-MAR-CA	Dress	Zara	Fashion
13-APR-CA	Toaster	Hamilton	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	Electronics
01-DEC-CA	Jeans	Levi's	Fashion
09-JUL-FR	Watch	Casio	Accessories

Find and Replace Dialog Box:

- Find what:** Electroni
- Replace with:** Electronics
- Within:** Sheet
- Search:** By Rows
- Look in:** Formulas
- Options:** Match case, Match entire cell contents (checked).
- Buttons:** Replace all, Replace, Find All, Find Next, Close.

Step 3: handling missing values

I filled the blank cells in the category column with “ unknown” to handle the missing values.

I filled the blank cells in the price column based on product name ,however some product names appeared only once so the average could not be calculated. Therefore , I used the category column as the criteria and calculated the average price.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Category	Price	Price
28-JAN-US	Laptop	Dell	1000	30	Electronics	Electronics	1000	1000
15-FEB-US	Sneakers	Nike	80	15	Fashion	Fashion	80	80
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Kitchen	130	130
11-APR-US	Smartphone	Samsung	900	25	Electronics	Electronics	900	900
22-MAY-US	Backpack	North Face	70	20	Unknown	Unknown	70	70
07-JUN-UK	Headphones	Sony		45	Electronics	Electronics	250	250
19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	Fashion	30	30
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Kitchen	90	90
05-SEP-UK	Tablet	Apple	500	50	Electronics	Electronics	500	500
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Outdoor	130	130
17-JUN-IN	Laptop	HP	950	25	Electronics	Electronics	950	950
25-NOV-AU	Sneakers	Adidas	90	40	Unknown	Unknown	90	90
08-DEC-DE	Coffee Maker	Nespresso	120	35	Unknown	Unknown	120	120
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Electronics	150	150
16-APR-ES	Headphones	Bose	250	20	Electronics	Electronics	250	250
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Accessories	50	50
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Electronics	160	160
27-JAN-IT	Laptop	Asus	980	10	Electronics	Electronics	980	980
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Fashion	150	150
14-AUG-US	Camping Tent	Coleman		10	Outdoor	Outdoor	#DIV/0!	130
14-MAY-RU	Camera	Nikon	700	50	Electronics	Electronics	700	700
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Kitchen	80	80
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Unknown	Unknown	150	150
29-SEP-CA	Smartphone	Google	800	45	Electronics	Electronics	800	800
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	Fashion	150	150
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Kitchen	400	400
07-MAR-CA	Dress	Zara	60	30	Fashion	Fashion	60	60
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Kitchen	40	40
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Electronics	130	130

Step 4: splitting and merging data

I used “left” and “right” functions to splitting the columns

Product ID	Product Name	Brand Name	Price	Quantity	Category	Manufacturing Date	Country Code	Product Brand
28-JAN-US	Laptop	Dell	1000	30	Electronics	28-JAN	US	Laptop Dell
15-FEB-US	Sneakers	Nike	80	15	Fashion	15-FEB	US	Sneakers Nike
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	03-MAR	US	Coffee Maker Keurig
11-APR-US	Smartphone	Samsung	900	25	Electronics	11-APR	US	Smartphone Samsung
22-MAY-US	Backpack	North Face	70	20	Unknown	22-MAY	US	Backpack North Face
07-JUN-UK	Headphones	Sony	250	45	Electronics	07-JUN	UK	Headphones Sony
19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	19-JUL	UK	T-Shirt Adidas
23-AUG-UK	Blender	Ninja	90	35	Kitchen	23-AUG	UK	Blender Ninja
10-SEP-UK	Tablet	Apple	500	50	Electronics	05-SEP	UK	Tablet Apple
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	14-OCT	UK	Hiking Boots Timberland
17-JUN-IN	Laptop	HP	950	25	Electronics	17-JUN	IN	Laptop HP
25-NOV-AU	Sneakers	Adidas	90	40	Unknown	25-NOV	AU	Sneakers Adidas
08-DEC-DE	Coffee Maker	Nespresso	120	35	Unknown	08-DEC	DE	Coffee Maker Nespresso
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	18-FEB	CA	Smartwatch Fitbit
16-APR-ES	Headphones	Bose	250	20	Electronics	16-APR	ES	Headphones Bose
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	21-AUG	CA	Laptop Bag Samsonite
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	20-AUG	CN	Smartwatch Huawei
27-JAN-IT	Laptop	Asus	980	10	Electronics	27-JAN	IT	Laptop Asus
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	01-MAR	UK	Sunglasses Oakley
14-AUG-US	Camping Tent	Coleman	130	10	Outdoor	14-AUG	US	Camping Tent Coleman
14-MAY-RU	Camera	Nikon	700	50	Electronics	14-MAY	RU	Camera Nikon
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	09-JAN	CA	Microwave Panasonic
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Unknown	19-JUL	BR	Fitness Tracker Xiaomi
29-SEP-CA	Smartphone	Google	800	45	Electronics	29-SEP	CA	Smartphone Google
03-JUN-CA	Sunglasses	Ray-Ban	150	25	Fashion	03-JUN	CA	Sunglasses Ray-Ban
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	11-JUL	CA	Blender Vitamix
07-MAR-CA	Dress	Zara	60	30	Fashion	07-MAR	CA	Dress Zara
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	13-APR	CA	Toaster Hamilton
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	24-MAY	CA	Fitness Tracker Garmin

Step 5: merging columns

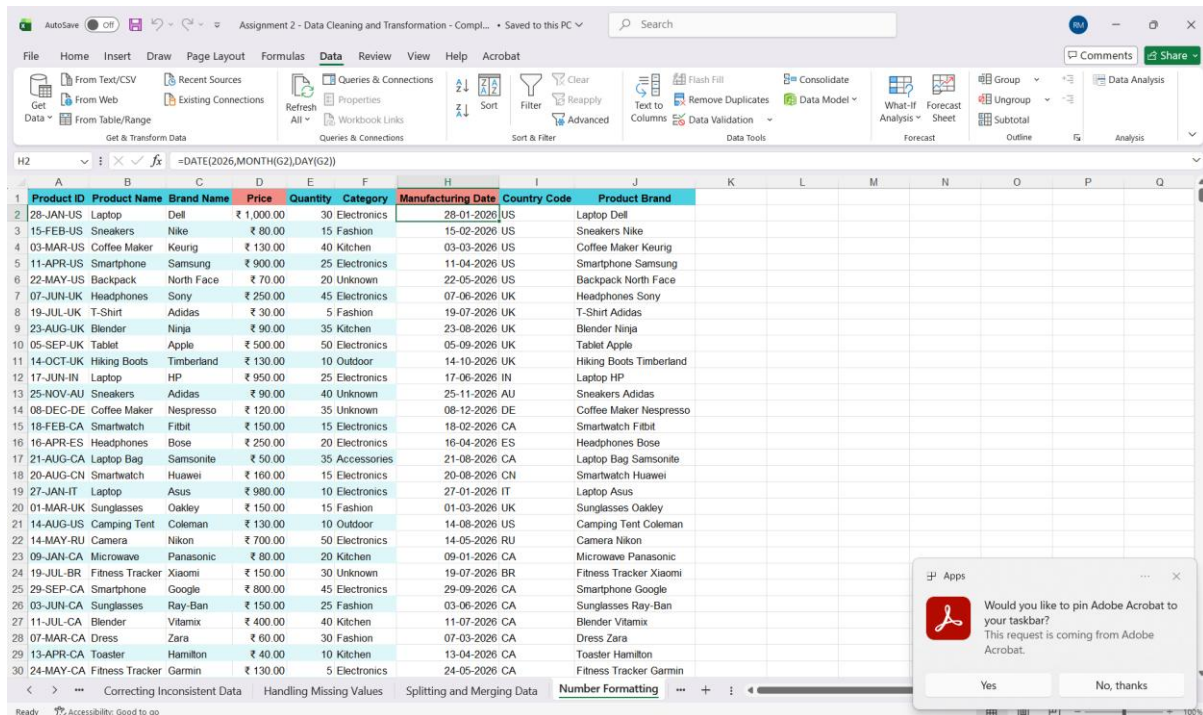
Merged the brand name and product name data using the formula

=B2&" "&C2

Product ID	Product Name	Brand Name	Price	Quantity	Category	Manufacturing Date	Country Code	Product Brand
28-JAN-US	Laptop	Dell	1000	30	Electronics	28-JAN	US	Laptop Dell
15-FEB-US	Sneakers	Nike	80	15	Fashion	15-FEB	US	Sneakers Nike
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	03-MAR	US	Coffee Maker Keurig
11-APR-US	Smartphone	Samsung	900	25	Electronics	11-APR	US	Smartphone Samsung
22-MAY-US	Backpack	North Face	70	20	Unknown	22-MAY	US	Backpack North Face
07-JUN-UK	Headphones	Sony	250	45	Electronics	07-JUN	UK	Headphones Sony
19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	19-JUL	UK	T-Shirt Adidas
23-AUG-UK	Blender	Ninja	90	35	Kitchen	23-AUG	UK	Blender Ninja
10-SEP-UK	Tablet	Apple	500	50	Electronics	05-SEP	UK	Tablet Apple
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	14-OCT	UK	Hiking Boots Timberland
17-JUN-IN	Laptop	HP	950	25	Electronics	17-JUN	IN	Laptop HP
25-NOV-AU	Sneakers	Adidas	90	40	Unknown	25-NOV	AU	Sneakers Adidas
08-DEC-DE	Coffee Maker	Nespresso	120	35	Unknown	08-DEC	DE	Coffee Maker Nespresso
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	18-FEB	CA	Smartwatch Fitbit
16-APR-ES	Headphones	Bose	250	20	Electronics	16-APR	ES	Headphones Bose
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	21-AUG	CA	Laptop Bag Samsonite
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	20-AUG	CN	Smartwatch Huawei
27-JAN-IT	Laptop	Asus	980	10	Electronics	27-JAN	IT	Laptop Asus
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	01-MAR	UK	Sunglasses Oakley
14-AUG-US	Camping Tent	Coleman	130	10	Outdoor	14-AUG	US	Camping Tent Coleman
14-MAY-RU	Camera	Nikon	700	50	Electronics	14-MAY	RU	Camera Nikon
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	09-JAN	CA	Microwave Panasonic
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Unknown	19-JUL	BR	Fitness Tracker Xiaomi
29-SEP-CA	Smartphone	Google	800	45	Electronics	29-SEP	CA	Smartphone Google
03-JUN-CA	Sunglasses	Ray-Ban	150	25	Fashion	03-JUN	CA	Sunglasses Ray-Ban
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	11-JUL	CA	Blender Vitamix
07-MAR-CA	Dress	Zara	60	30	Fashion	07-MAR	CA	Dress Zara
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	13-APR	CA	Toaster Hamilton
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	24-MAY	CA	Fitness Tracker Garmin

Step 6:

Number formatting



Assignment 2 - Data Cleaning and Transformation - Compl... Saved to this PC

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat

Get Data: From Text/CSV, From Web, From Table/Range, Recent Sources, Existing Connections

Queries & Connections: Refresh All, Properties, Workbook Links

Sort & Filter: Sort, Filter, Clear, Reapply, Advanced

Data Tools: Flash Fill, Remove Duplicates, Consolidate, Data Model, Text to Columns, Data Validation

Forecast: What-If Analysis, Forecast Sheet

Outline: Group, Ungroup, Subtotal, Data Analysis

Formula Bar: H2, =DATE(2026,MONTH(G2),DAY(G2))

Product ID	Product Name	Brand Name	Price	Quantity	Category	Manufacturing Date	Country Code	Product Brand
28-JAN-US	Laptop	Dell	₹ 1,000.00	30	Electronics	28-01-2026	US	Laptop Dell
15-FEB-US	Sneakers	Nike	₹ 80.00	15	Fashion	15-02-2026	US	Sneakers Nike
03-MAR-US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	03-03-2026	US	Coffee Maker Keurig
11-APR-US	Smartphone	Samsung	₹ 900.00	25	Electronics	11-04-2026	US	Smartphone Samsung
22-MAY-US	Backpack	North Face	₹ 70.00	20	Unknown	22-05-2026	US	Backpack North Face
07-JUN-UK	Headphones	Sony	₹ 250.00	45	Electronics	07-06-2026	UK	Headphones Sony
19-JUL-UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	19-07-2026	UK	T-Shirt Adidas
23-AUG-UK	Blender	Ninja	₹ 90.00	35	Kitchen	23-08-2026	UK	Blender Ninja
10-SEP-UK	Tablet	Apple	₹ 500.00	50	Electronics	05-09-2026	UK	Tablet Apple
14-OCT-UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	14-10-2026	UK	Hiking Boots Timberland
17-JUN-IN	Laptop	HP	₹ 950.00	25	Electronics	17-06-2026	IN	Laptop HP
25-NOV-AU	Sneakers	Adidas	₹ 90.00	40	Unknown	25-11-2026	AU	Sneakers Adidas
08-DEC-DE	Coffee Maker	Nespresso	₹ 120.00	35	Unknown	08-12-2026	DE	Coffee Maker Nespresso
18-FEB-CA	Smartwatch	Fitbit	₹ 150.00	15	Electronics	18-02-2026	CA	Smartwatch Fitbit
16-APR-ES	Headphones	Bose	₹ 250.00	20	Electronics	16-04-2026	ES	Headphones Bose
21-AUG-CA	Laptop Bag	Samsontite	₹ 50.00	35	Accessories	21-08-2026	CA	Laptop Bag Samsontite
20-AUG-CN	Smartwatch	Huawei	₹ 160.00	15	Electronics	20-08-2026	CN	Smartwatch Huawei
27-JAN-IT	Laptop	Asus	₹ 980.00	10	Electronics	27-01-2026	IT	Laptop Asus
01-MAR-UK	Sunglasses	Oakley	₹ 150.00	15	Fashion	01-03-2026	UK	Sunglasses Oakley
14-AUG-US	Camping Tent	Coleman	₹ 130.00	10	Outdoor	14-08-2026	US	Camping Tent Coleman
14-MAY-RU	Camera	Nikon	₹ 700.00	50	Electronics	14-05-2026	RU	Camera Nikon
09-JAN-CA	Microwave	Panasonic	₹ 80.00	20	Kitchen	09-01-2026	CA	Microwave Panasonic
19-JUL-BR	Fitness Tracker	Xiaomi	₹ 150.00	30	Unknown	19-07-2026	BR	Fitness Tracker Xiaomi
29-SEP-CA	Smartphone	Google	₹ 800.00	45	Electronics	29-09-2026	CA	Smartphone Google
03-JUN-CA	Sunglasses	Ray-Ban	₹ 150.00	25	Fashion	03-06-2026	CA	Sunglasses Ray-Ban
11-JUL-CA	Blender	Vitamix	₹ 400.00	40	Kitchen	11-07-2026	CA	Blender Vitamix
07-MAR-CA	Dress	Zara	₹ 60.00	30	Fashion	07-03-2026	CA	Dress Zara
13-APR-CA	Toaster	Hamilton	₹ 40.00	10	Kitchen	13-04-2026	CA	Toaster Hamilton
24-MAY-CA	Fitness Tracker	Garmin	₹ 130.00	5	Electronics	24-05-2026	CA	Fitness Tracker Garmin

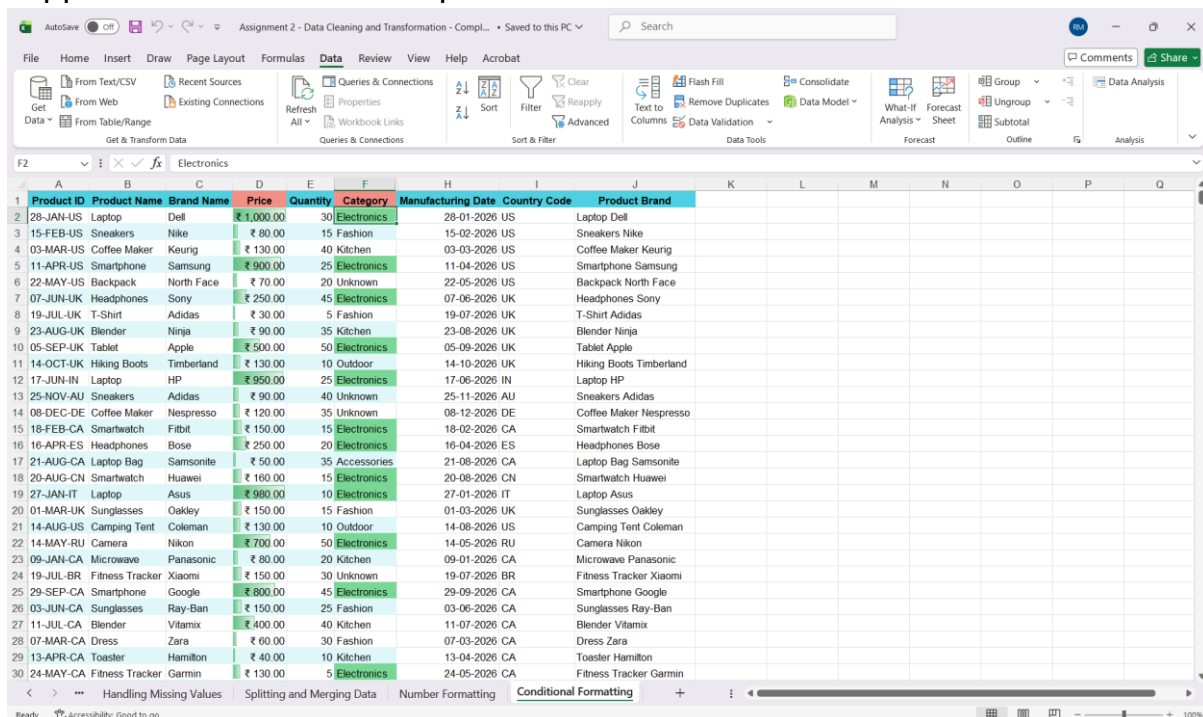
Ready Accessibility: Good to go

Apps: Would you like to pin Adobe Acrobat to your taskbar? This request is coming from Adobe Acrobat. Yes No, thanks

Step 7: conditional formatting

I applied the conditional formatting to the category to highlight all cells containing electronics

I applied the data bar to the price column .



Assignment 2 - Data Cleaning and Transformation - Compl... Saved to this PC

File Home Insert Draw Page Layout Formulas Data Review View Help Acrobat

Get Data: From Text/CSV, From Web, From Table/Range, Recent Sources, Existing Connections

Queries & Connections: Refresh All, Properties, Workbook Links

Sort & Filter: Sort, Filter, Clear, Reapply, Advanced

Data Tools: Flash Fill, Remove Duplicates, Consolidate, Data Model, Text to Columns, Data Validation

Forecast: What-If Analysis, Forecast Sheet

Outline: Group, Ungroup, Subtotal, Data Analysis

Formula Bar: F2, Electronics

Product ID	Product Name	Brand Name	Price	Quantity	Category	Manufacturing Date	Country Code	Product Brand
28-JAN-US	Laptop	Dell	₹ 1,000.00	30	Electronics	28-01-2026	US	Laptop Dell
15-FEB-US	Sneakers	Nike	₹ 80.00	15	Fashion	15-02-2026	US	Sneakers Nike
03-MAR-US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	03-03-2026	US	Coffee Maker Keurig
11-APR-US	Smartphone	Samsung	₹ 900.00	25	Electronics	11-04-2026	US	Smartphone Samsung
22-MAY-US	Backpack	North Face	₹ 70.00	20	Unknown	22-05-2026	US	Backpack North Face
07-JUN-UK	Headphones	Sony	₹ 250.00	45	Electronics	07-06-2026	UK	Headphones Sony
19-JUL-UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	19-07-2026	UK	T-Shirt Adidas
23-AUG-UK	Blender	Ninja	₹ 90.00	35	Kitchen	23-08-2026	UK	Blender Ninja
10-SEP-UK	Tablet	Apple	₹ 500.00	50	Electronics	05-09-2026	UK	Tablet Apple
14-OCT-UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	14-10-2026	UK	Hiking Boots Timberland
17-JUN-IN	Laptop	HP	₹ 950.00	25	Electronics	17-06-2026	IN	Laptop HP
25-NOV-AU	Sneakers	Adidas	₹ 90.00	40	Unknown	25-11-2026	AU	Sneakers Adidas
08-DEC-DE	Coffee Maker	Nespresso	₹ 120.00	35	Unknown	08-12-2026	DE	Coffee Maker Nespresso
18-FEB-CA	Smartwatch	Fitbit	₹ 150.00	15	Electronics	18-02-2026	CA	Smartwatch Fitbit
16-APR-ES	Headphones	Bose	₹ 250.00	20	Electronics	16-04-2026	ES	Headphones Bose
21-AUG-CA	Laptop Bag	Samsontite	₹ 50.00	35	Accessories	21-08-2026	CA	Laptop Bag Samsontite
20-AUG-CN	Smartwatch	Huawei	₹ 160.00	15	Electronics	20-08-2026	CN	Smartwatch Huawei
27-JAN-IT	Laptop	Asus	₹ 980.00	10	Electronics	27-01-2026	IT	Laptop Asus
01-MAR-UK	Sunglasses	Oakley	₹ 150.00	15	Fashion	01-03-2026	UK	Sunglasses Oakley
14-AUG-US	Camping Tent	Coleman	₹ 130.00	10	Outdoor	14-08-2026	US	Camping Tent Coleman
14-MAY-RU	Camera	Nikon	₹ 700.00	50	Electronics	14-05-2026	RU	Camera Nikon
09-JAN-CA	Microwave	Panasonic	₹ 80.00	20	Kitchen	09-01-2026	CA	Microwave Panasonic
19-JUL-BR	Fitness Tracker	Xiaomi	₹ 150.00	30	Unknown	19-07-2026	BR	Fitness Tracker Xiaomi
29-SEP-CA	Smartphone	Google	₹ 800.00	45	Electronics	29-09-2026	CA	Smartphone Google
03-JUN-CA	Sunglasses	Ray-Ban	₹ 150.00	25	Fashion	03-06-2026	CA	Sunglasses Ray-Ban
11-JUL-CA	Blender	Vitamix	₹ 400.00	40	Kitchen	11-07-2026	CA	Blender Vitamix
07-MAR-CA	Dress	Zara	₹ 60.00	30	Fashion	07-03-2026	CA	Dress Zara
13-APR-CA	Toaster	Hamilton	₹ 40.00	10	Kitchen	13-04-2026	CA	Toaster Hamilton
24-MAY-CA	Fitness Tracker	Garmin	₹ 130.00	5	Electronics	24-05-2026	CA	Fitness Tracker Garmin

Ready Accessibility: Good to go

Conditional Formatting